

Title: Eco Drive Motors: Data Analytics & Sustainability Report

1.0 Executive Summary

This report presents a comprehensive data analysis conducted for Eco Drive Motors, a UK-based automotive company navigating the transition towards sustainable mobility in line with the UK government's 2030 ban on new petrol and diesel vehicles. Utilizing Power BI, we developed two interactive dashboards to analyze key performance indicators (KPIs) across sustainability, sales revenue, cost analysis, and customer segmentation. The analysis reveals that while Electric Vehicles (EVs) command the highest average selling price (£32.54K overall average, with EVs being the most expensive type) and are crucial for long-term sustainability, they currently represent only 18.45% of the sales mix (Q4 2023). Petrol vehicles dominate sales (61.69%), indicating a significant transition gap. Top customers exhibit remarkable loyalty, with the top 10 accounting for 8,942 units. Strategic recommendations are provided to accelerate EV adoption, optimize revenue, and enhance customer loyalty to meet 2030 targets profitably.

2.0 Introduction & Project Objectives

2.1 Background

The UK's legislated path to net-zero by 2050, with a pivotal 2030 milestone, fundamentally disrupts the automotive sector. Eco Drive Motors, selling Diesel, Petrol, and Electric vehicles, must balance current profitability with a mandatory strategic pivot. Data-driven decision-making is no longer optional but essential for survival and growth.

2.2 Objective

To transform raw sales data into actionable strategic insights using Power BI, focusing on four core business dimensions:

1. Sustainability Progress: Tracking EV adoption against the 2030 target.
2. Financial Performance: Understanding revenue streams and pricing.
3. Cost Structures: Analyzing the Total Cost of Ownership (TCO) components.

4. Customer Base: Identifying high-value segments and loyalty patterns.

3.0 Dashboard Analysis & KPI Rationale

We designed two focused dashboards, each containing two primary KPIs for clarity and strategic alignment.

3.1 Dashboard 1: Sustainability & Sales Performance

- Primary Focus: Macro-level progress and sales composition.

1. KPI 1: Electric Vehicle Market Share (Sustainability)

- Visual: Gauge chart showing 18.45%.
- Rationale: This is the single most critical metric for tracking compliance with the 2030 mandate. It directly measures the company's transition speed. The current 18.45% share highlights the scale of the challenge, indicating that over 80% of sales are still internal combustion engine (ICE) vehicles.
- Insight: The dashboard includes an interactive date slicer (1/1/2023 - 12/31/2023), allowing trend analysis over time to see if EV share is growing quarter-by-quarter.

2. KPI 2: Vehicle Sales Distribution & Revenue by Type (Sales Revenue)

- Visuals: Donut chart and bar chart.
- Rationale: The donut chart breaks down the sales mix: Petrol (61.69%), Diesel (19.86%), Electric (18.45%). This reveals operational and inventory focus. The revenue bar chart shows the average revenue contribution per vehicle type, crucial for understanding profitability.
- Insight: A critical disconnect is observed: while EV sales volume is lowest, the revenue per EV is the highest. This suggests EVs are premium products driving higher revenue per unit, a key factor for financial planning during the transition.

3.2 Dashboard 2: Cost & Customer Analysis

- Primary Focus: Micro-level pricing strategies and client loyalty.
1. KPI 3: Average Selling Price & Price by Vehicle Type (Total Cost of Ownership - Selling Price Component)
 - Visuals: Card visual (£32.54K avg) and column chart.
 - Rationale: The overall Average Selling Price is a fundamental health metric. Disaggregating this by vehicle type (column chart) is essential for pricing strategy, margin analysis, and customer targeting. It forms a core part of the TCO discussion for customers.
 - Insight: The column chart visually confirms EVs as the premium segment. Understanding this price differential is vital for developing marketing strategies, financing options, and potential subsidies to offset the higher upfront cost for buyers.
 2. KPI 4: Top 10 Customers by Units Purchased (Customer Segmentation)
 - Visual: Table of top customers.
 - Rationale: Identifies the most valuable B2B or fleet clients by volume, which is crucial for retention and understanding B2B vs. B2C dynamics. Loyalty here indicates satisfaction and potential for EV fleet conversions.
 - Insight: The top 10 customers are responsible for a staggering 8,942 vehicle units, indicating a heavy reliance on a few large clients (e.g., Liam Johnson at 933 units). This presents both a risk (concentration) and an opportunity (targeted EV transition programs for fleets).

4.0 Key Findings & Integrated Insights

1. The Sustainability-Revenue Paradox: EVs, crucial for the company's future, have the lowest sales volume (18.45%) but generate the highest revenue per unit. The strategy must bridge this volume gap without eroding the premium price point.

2. **Dominance of ICE Vehicles:** Petrol and diesel combined still constitute over 80% of sales. A managed, strategic decline of this segment is needed while ramping up EV capacity and appeal.
3. **Customer Concentration Power:** The sales model is heavily reliant on a small cohort of high-volume buyers. Their purchasing decisions will disproportionately impact the ability to hit 2030 EV targets.
4. **Data-Driven Pricing Confidence:** The clear pricing hierarchy (EV > Petrol > Diesel) validates the current pricing model and provides a benchmark for future adjustments and promotions.

5.0 Strategic Recommendations for the CEO

Based on the dashboard insights, we recommend the following actionable strategies:

1. **Launch a "Fleet-First" EV Transition Programme:**
 - **Action:** Proactively engage the Top 10 customers (e.g., Liam Johnson, Ethan Young) with tailored EV fleet packages. Offer cost-benefit analyses showcasing TCO savings, dedicated account management, and bulk purchase discounts.
 - **Rationale:** Converting these high-volume buyers will rapidly increase EV market share and provide stable, large-scale demand. This mitigates concentration risk by deepening relationships.
2. **Implement Targeted EV Incentives to Boost Volume:**
 - **Action:** Develop targeted promotions for the retail (B2C) segment without devaluing the brand. This could include subsidised charging points, extended warranty packages, or competitive finance rates highlighting lower running costs.
 - **Rationale:** To move from 18.45% towards a majority EV share, the upfront price barrier must be addressed. Strategic incentives can stimulate volume while protecting margin better than across-the-board price cuts.
3. **Develop a Phased-Out Plan for ICE Vehicles:**

- Action: Use the date slicer to forecast the decline of petrol/diesel sales. Align inventory procurement, marketing spends, and dealership training with a gradual phase-out, reallocating resources to EV sales and service.
- Rationale: Prevents overstocking of soon-to-be-obsolete models and smoothly reallocates capital and human resources towards the future product line.

4. Enhance Dashboards for Deeper TCO Analysis:

- Action: Integrate data on resale value and maintenance costs by vehicle type into the Cost Analysis dashboard.
- Rationale: A full TCO visualization (Selling Price + Maintenance - Resale Value) would be a powerful sales tool for EVs, demonstrating long-term savings to counteract higher sticker prices.

6.0 Conclusion

The Power BI dashboards have successfully transformed Eco Drive Motors' sales data into a clear strategic map. The path to 2030 is challenging but clear: the company must leverage its strong customer relationships and premium EV positioning to drive volume. The dual focus on converting high-value fleets and making EVs more accessible to retail customers, guided by continuous interactive data analysis, will be critical. By acting on these recommendations, Eco Drive Motors can align profitability with regulatory mandates, securing its role as a leader in the UK's sustainable automotive future.

7.0 References

1. UK Government (2020). *The Ten Point Plan for a Green Industrial Revolution*. Retrieved from gov.uk
2. Porter, M. E., & Heppelmann, J. E. (2015). *How Smart, Connected Products Are Transforming Companies*. Harvard Business Review.
3. Microsoft Power BI Documentation. (2024). *Guide to Data Visualization*. Retrieved from learn.microsoft.com
4. The Society of Motor Manufacturers and Traders (SMMT). (2023). *UK Automotive Sustainability Report*.